

Machine Learning Fundamentals I: Basic Concepts

Machine Learning Fundamentals II: Model Training and Evaluation

● Introduction to Module 8's Study
Video • 1 min

● Introduction to Module 8's Study
Video • 1 min

What Does “Good” Mean in Machine Learning Models?
Reading • 5 min

ML Model Training Fundamentals: Parameters, Hyperparameters, and Loss Functions
Video • 8 min

ML Model Training Fundamentals: Parameters, Hyperparameters, and Loss Functions
Video • 8 min

ML Model Training Fundamentals: Loss Functions and Gradient Descent
Video • 5 min

ML Model Training Fundamentals: Gradient Descent

Video • 10 min

✔ Materials for ML model training
Reading • 10 min

What Does It Actually Mean to “Train” a Machine Learning Model?

Reading • 5 min

ML Model Evaluation Fundamentals:
 Fundamental Concepts in ML Model Evaluation
 Video • 13 min

ML Model Evaluation Fundamentals:
 Fundamental Concepts in ML Model Evaluation
 Video • 13 min

ML Model Evaluation Fundamentals:
 Fundamental Concepts in ML Model Evaluation
 Video • 5 min

Materials on ML Model Evaluation

✓ Fundamentals

Reading • 10 min

When a Model Fails: Is It Bias, Variance, or Something Else?
Reading • 5 min

Summary of the ML Process
Video • 5 min

Summary of the ML Process
Video • 5 min

Materials for ML Process
Reading • 10 min

From Data to Decisions: Reconstructing the Machine Learning Pipeline
Reading • 5 min

Module 8 Learning Coach
Dialogue • 5 min

Module 8 Quiz
Practice Assignment • 30 min

Module 8 Assignment 1
Graded App Item • 30 min

Module 8 Assignment 1
Graded App Item • 30 min

Module 8 Assignment 2
Graded App Item • 30 min

CSV file for Module 8 Assignment
Reading • 1 min

Module 8 Recap
Reading • 5 min

Data Preparation

Support Vector Machines

Artificial Neural Networks

00

0:01 / 1:30

 $2x$ 

- Reflect on the complete machine learning workflow in your own words, beginning from raw data and data representation, moving through model selection and training, and ending with evaluation and generalization.
- Consider which stage of the workflow you believe is the most fragile or error-prone in practice. Reflect on why mistakes at this stage could significantly affect the final model.

- Reflect on how poor data representation or preprocessing decisions could limit model performance, even if model training and evaluation are performed correctly.
- Consider how an inappropriate evaluation strategy could create false confidence in a model that may not actually generalize well to real-world data.

- Parametric vs. non-parametric models
- Bias vs. variance
- Training longer vs. improving generalization
- Reflect on which trade-off you currently find the most difficult to reason about and explain why.
- Consider any remaining questions or uncertainties you still have regarding these trade-offs and their practical implications.

- Reflect on the areas of machine learning where you now feel more confident after completing these topics.
- Consider which parts of the machine learning process still feel unclear, difficult to interpret, or like a “black box” to you, and explain why.

Mark as completed

Like Dislike Report an issue

Go to next item →